

(KTXFI2EBNF) Physics II. Test 1

December 1, 2025

1. An electron has a kinetic energy of 5 eV.
 - (a) What is its velocity?
 - (b) What is its momentum?
 - (c) What is the wavelength of the associated de Broglie wave? (10 points)

2. A metal surface is illuminated with light of wavelength $1.1 \cdot 10^{-7}$ m. What is the velocity of the emitted electrons if the photoelectric effect starts at a wavelength of $2.77 \cdot 10^{-7}$ m?
($m_e = 9.1 \cdot 10^{-31}$ kg) (10 points)

3. In a Compton scattering experiment, X-rays with a wavelength of 0.1 nm are used.
($m_e = 9.1 \cdot 10^{-31}$ kg)
 - (a) At what scattering angle does the wavelength of the radiation increase by 1%?
 - (b) At what angle does the wavelength become 0.01% larger? (10 points)

4. A light source has a power output of 100 W.
 - (a) What is the frequency of the light if the entire energy of the source is emitted as light with wavelength $\lambda = 400$ nm?
 - (b) How many photons are emitted by the light source per second? (10 points)

5. We study a positron in a radioactive particle beam as it passes through the laboratory. It is found that, as measured in the laboratory, the average lifetime of the particle is 50 ns; after this time the particle decays. The average lifetime measured in the reference frame attached to the particles is 7.5 ns. What is the propagation velocity of the particle beam? Determine the value of $\frac{m}{m_0}$ for the electron! Calculate the rest energy of a positron corresponding to its rest mass ($m_p = 9.1 \cdot 10^{-31}$ kg) in joules and in electronvolts!
(1 MeV = $1.6 \cdot 10^{-13}$ J) (10 points)

Condition for obtaining a signature: achieving at least 40% (20 points) on the midterm exam.

Grades: 0–25 points: Fail (1), 26–30 points: Pass (2), 31–37 points: Satisfactory (3), 38–45 points: Good (4), 46–50 points: Excellent (5).

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